

Mechanisation

Land Sciences

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Short Title: Mechanisation
Faculty Engineering
Department Land Sciences
Cluster Engineering
Author TWOODCOCK
Credits 5

Level: Introductory

Description of Module / Aims

This module will enable the student to acquire a comprehensive knowledge of tractor technology and a detailed understanding of the selection, operation and maintenance of crop establishment, organic manure handling, harvesting and forage production and feeding equipment.

Programmes

MECH-0007 BSc in Agriculture (WD_SAGUL_D)

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Indicative Content

- Tractor technologies
- Electronics and precision control systems
- Crop establishment equipment
- Organic manure handling equipment
- Harvesting equipment
- Forage production and feeding equipment
- Workshop tools

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe modern tractors systems including their component design.
2. Explain electronic and precision control systems on modern machinery.
3. Describe crop establishment equipment.
4. Discuss organic manure handling equipment.
5. Describe harvesting equipment and systems.
6. Demonstrate an understanding of the machines involved in forage production and feeding.
7. Identify workshop hand tools and power tools.

Learning and Teaching Methods

- Lectures
- Self-directed learning
- Practical demonstrations

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6,7
Continuous Assessment	50%	2,3,4,6,7
<i>Practical</i>	50%	2,3,4,6,7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter

40%--49%: Will be able to show a basic knowledge of the key learning outcomes

50%--59%: As well as the above the student will be able to demonstrate an understanding of some of the complexities of the topics

60%--69%: In addition, the student will demonstrate more detailed knowledge of the all topics including and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material.

70%--100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learn

Note: In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Teagasc. _Organic Manure Handling._ Dublin. 2020.
- Teagasc. _Workshop skills_ Dublin. 2019.

Requested Resources

- WORKSHOP: Kildalton Agricultural College Equipment and Machinery